로지스틱 회귀와 분류(3)

다층신경망과 비선형 결정경계

제주대학교 컴퓨터공학과
변영철 교수

https://github.com/yungbyun/ml

학습과 결정경계

train 객체
경사하강(3, 4, 5) 해주는!

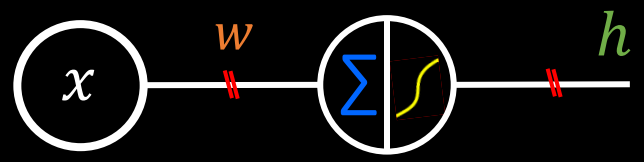
대충 만들어진 초기 결정경계

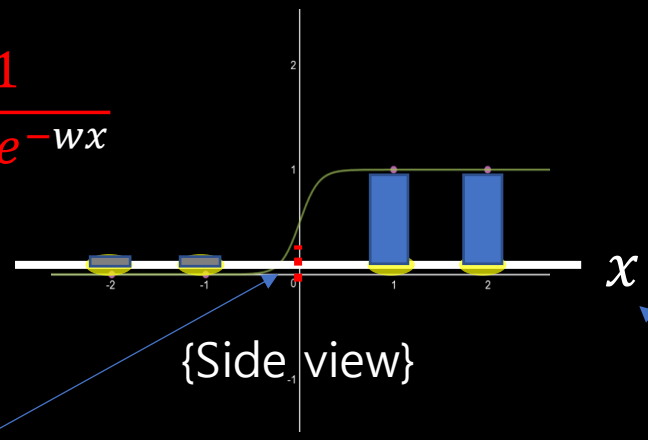
1. 파라미터 난수 초기화 (w, b)
2. 오류 함수 E 계산 그래프 생성
3. 앞으로 전파
4. 역전파 및 w, b 변화가 오류에 (기)미치는 영향 (체인룰)
5. 경사 하강
6. (goto 3) 반복적으로 w, b 업데이트
7. 학습 완료 후 w, b

잘 구분하는, 제대로 된 결정경계

1-Input Neuron

$$h = \frac{1}{1 + e^{-wx}}$$





{Side view}

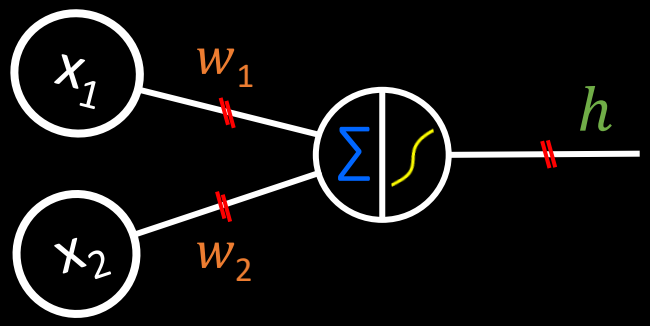
입력 1개

Decision boundary : 직선 상의 점

$$x \cdot w = 0$$
$$x = 0$$

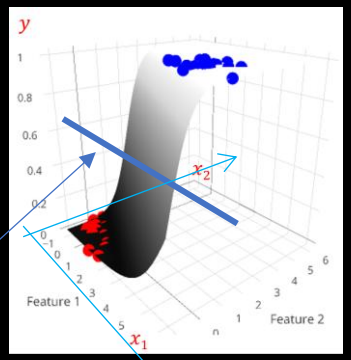
{View from above}

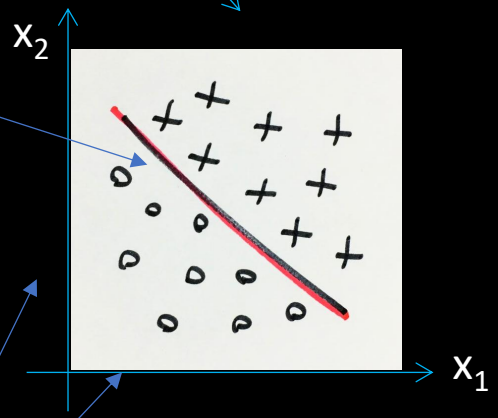
2-Input Neuron



Decision boundary : 평면 상의 직선(line)

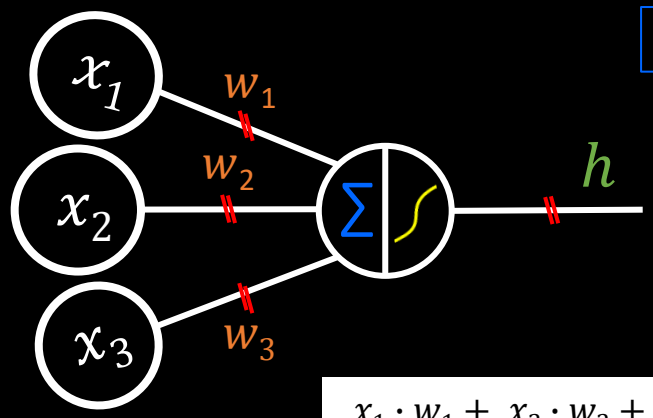
$$x_1 \cdot w_1 + x_2 \cdot w_2 = 0$$





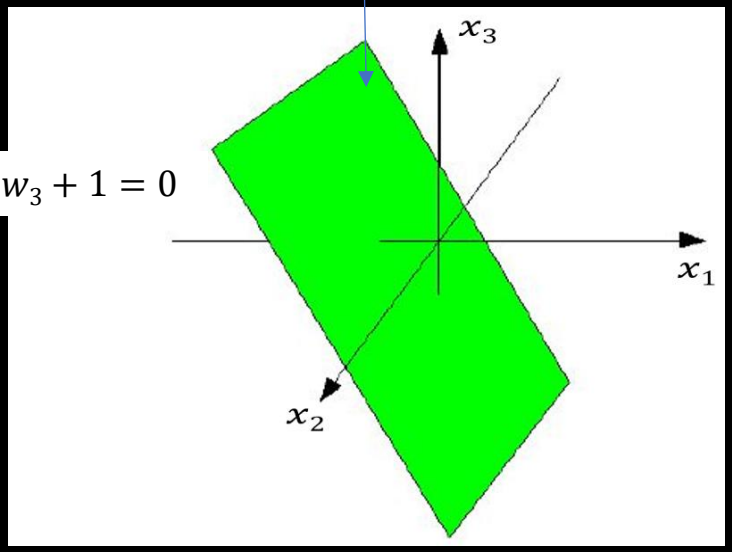
입력 2개

3-Input Neuron



Decision boundary : 공간 상의 평면(plane)

$$x_1 \cdot w_1 + x_2 \cdot w_2 + x_3 \cdot w_3 + 1 = 0$$



$$x_1 \cdot w_1 + x_2 \cdot w_2 + x_3 \cdot w_3 + x_4 \cdot w_4 = 0$$

4입력이 이상이면
→ 초평면(hyperplane)

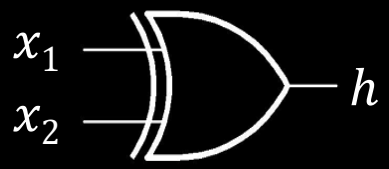
이제까지는 모두 선형 결정경계로 패턴을 분류

점, 선, 평면, 초평면 결정경계

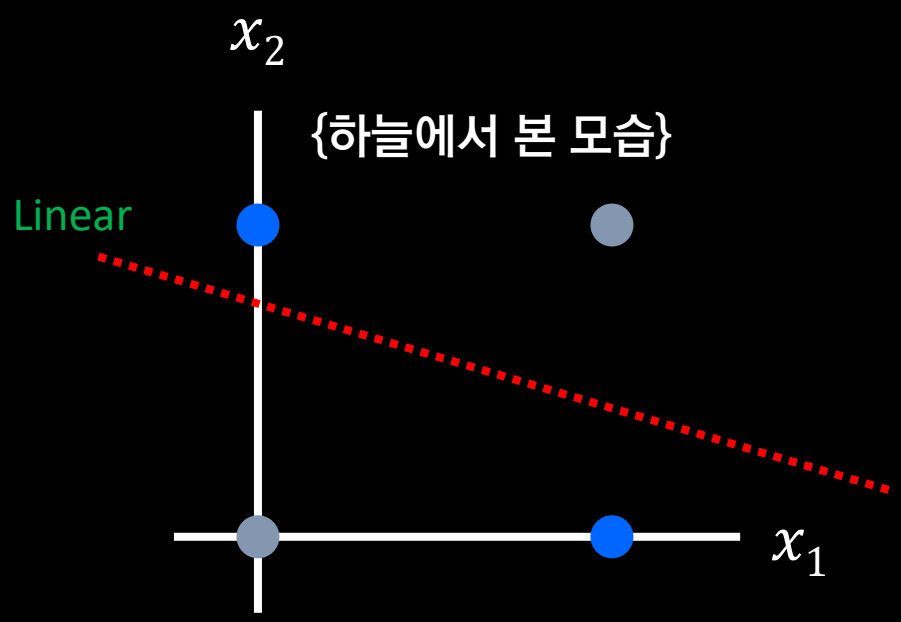
- 선형 모델? 비선형 모델?
- 신경세포의 대답(hypothesis)이 계수(coefficient)들의 선형 결합이면 선형모델

$$h = w_1 \cdot x_1 + w_2 \cdot x_2$$

XOR 문제



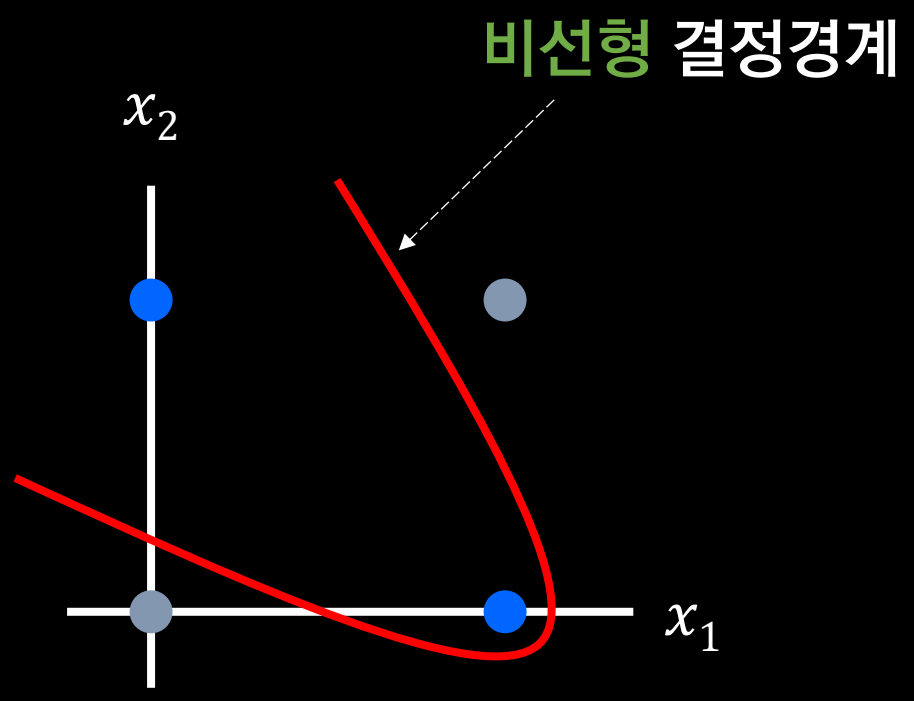
x_1	x_2	h
0	0	0
0	1	1
1	0	0
1	1	1



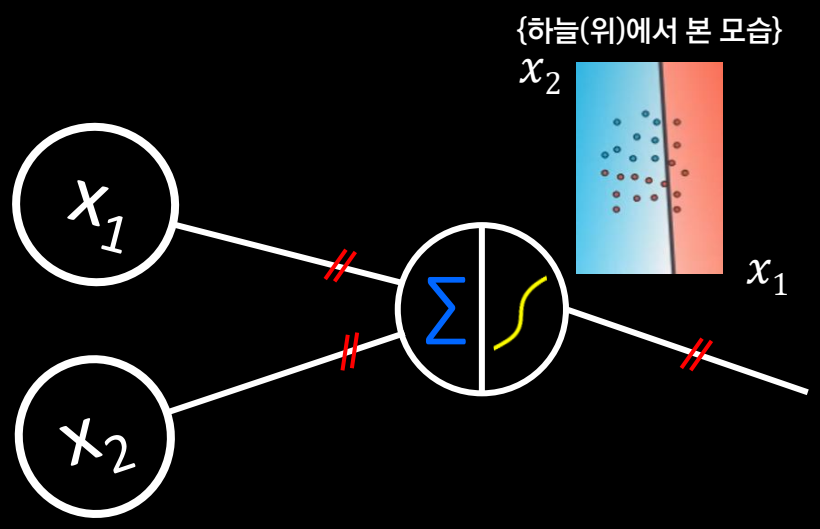
- 신경세포의 출력 h는 0 혹은 1
- 따라서 필요한 결정경계의 수는?
- 선형 결정경계 1개로는 불가능

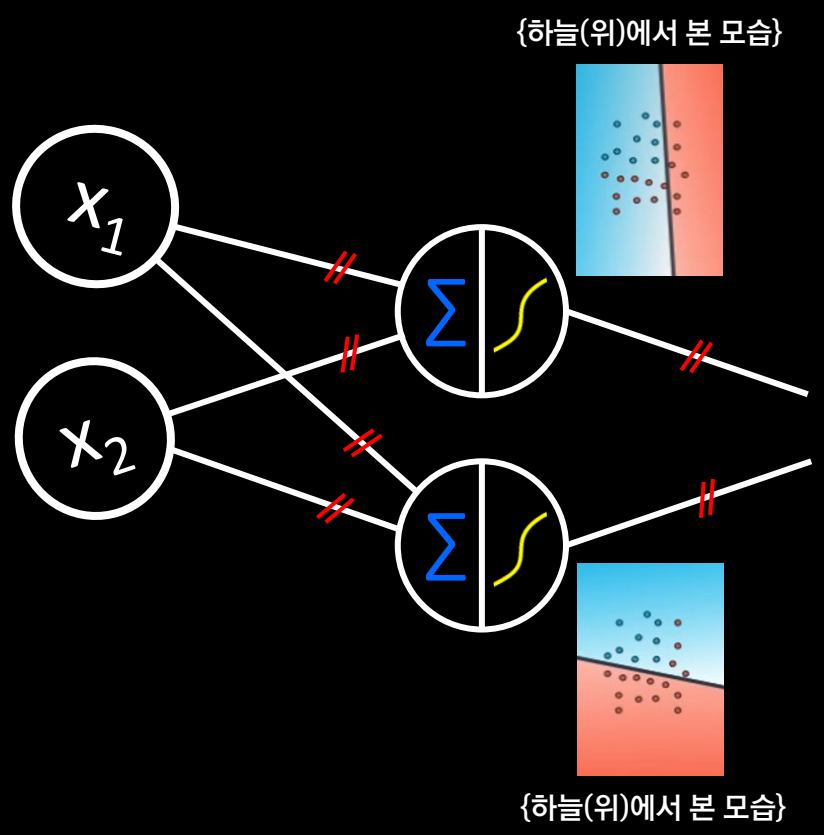
(실습) 16.py

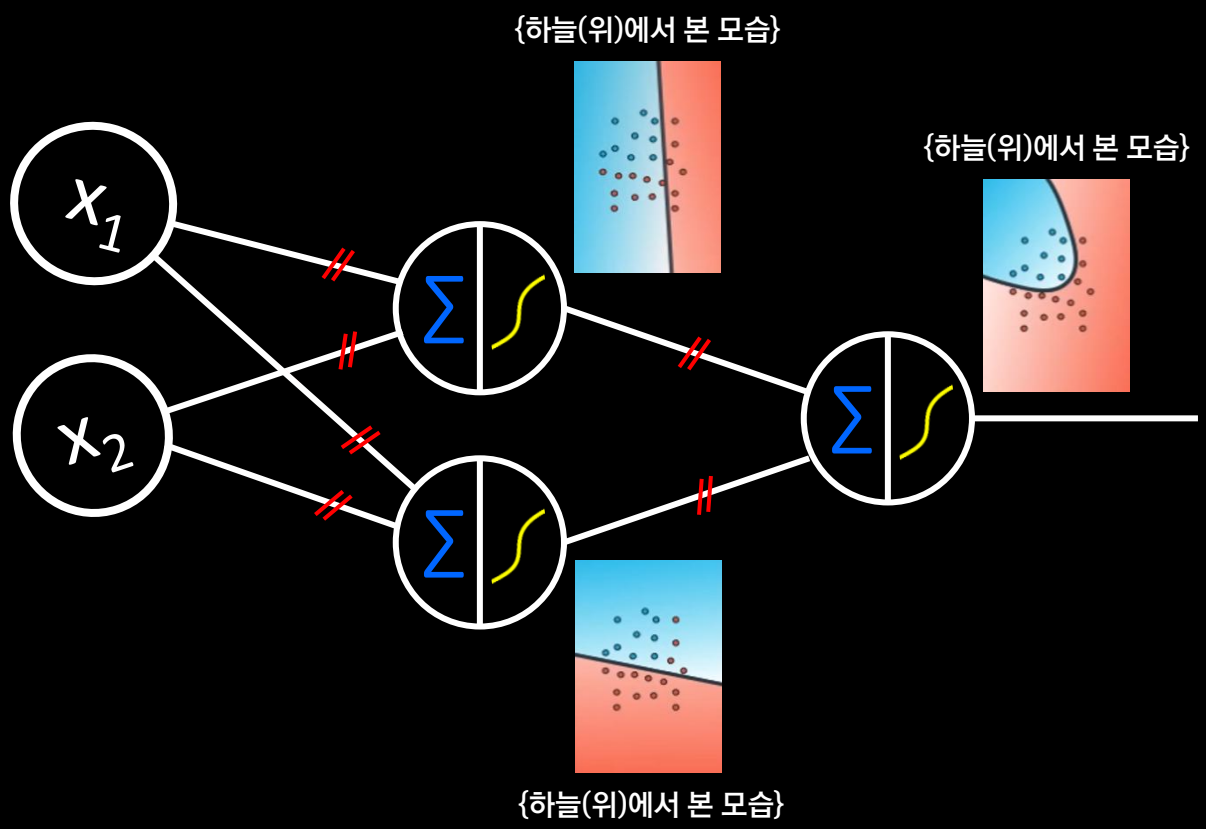
- 신경세포 하나
- 선형 결정 경계 1개
- 해결 불가능



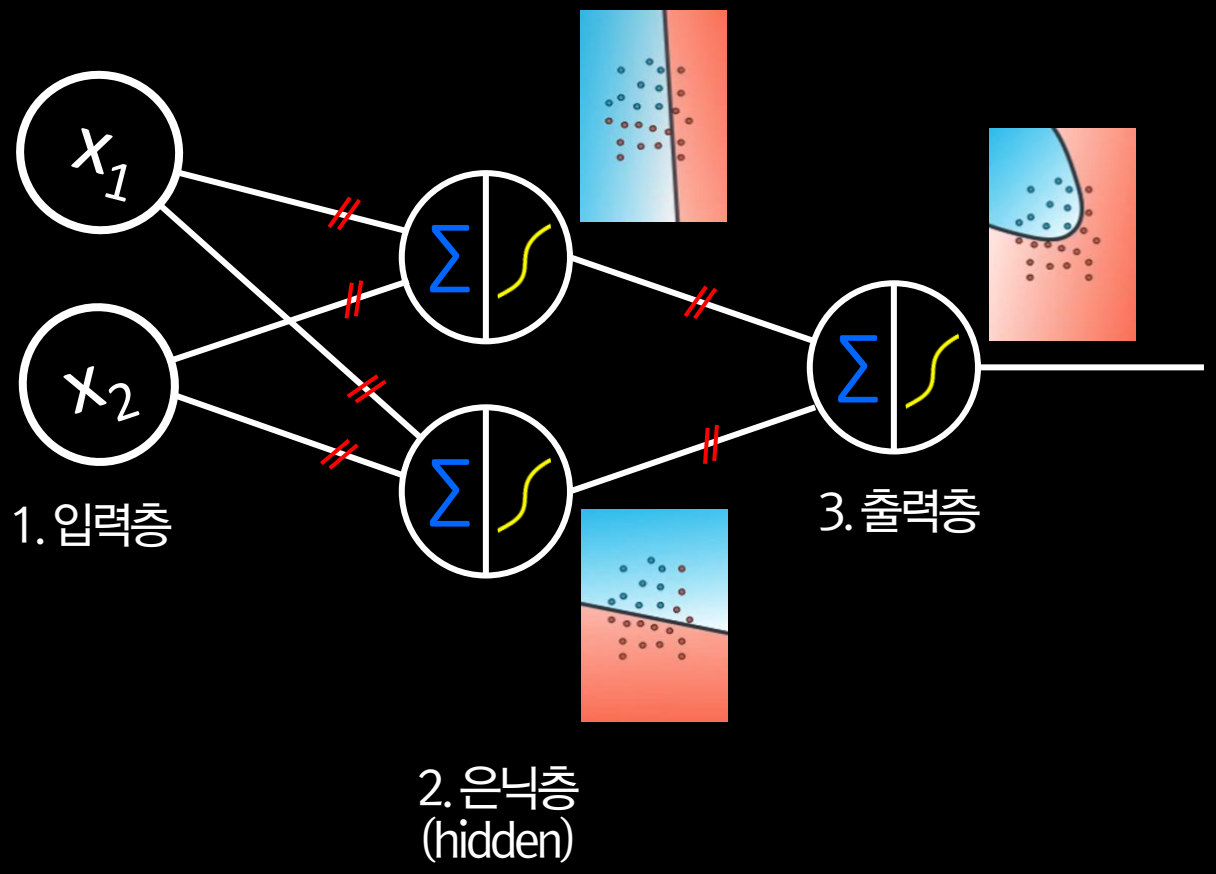
{하늘(위)에서 본 모습}

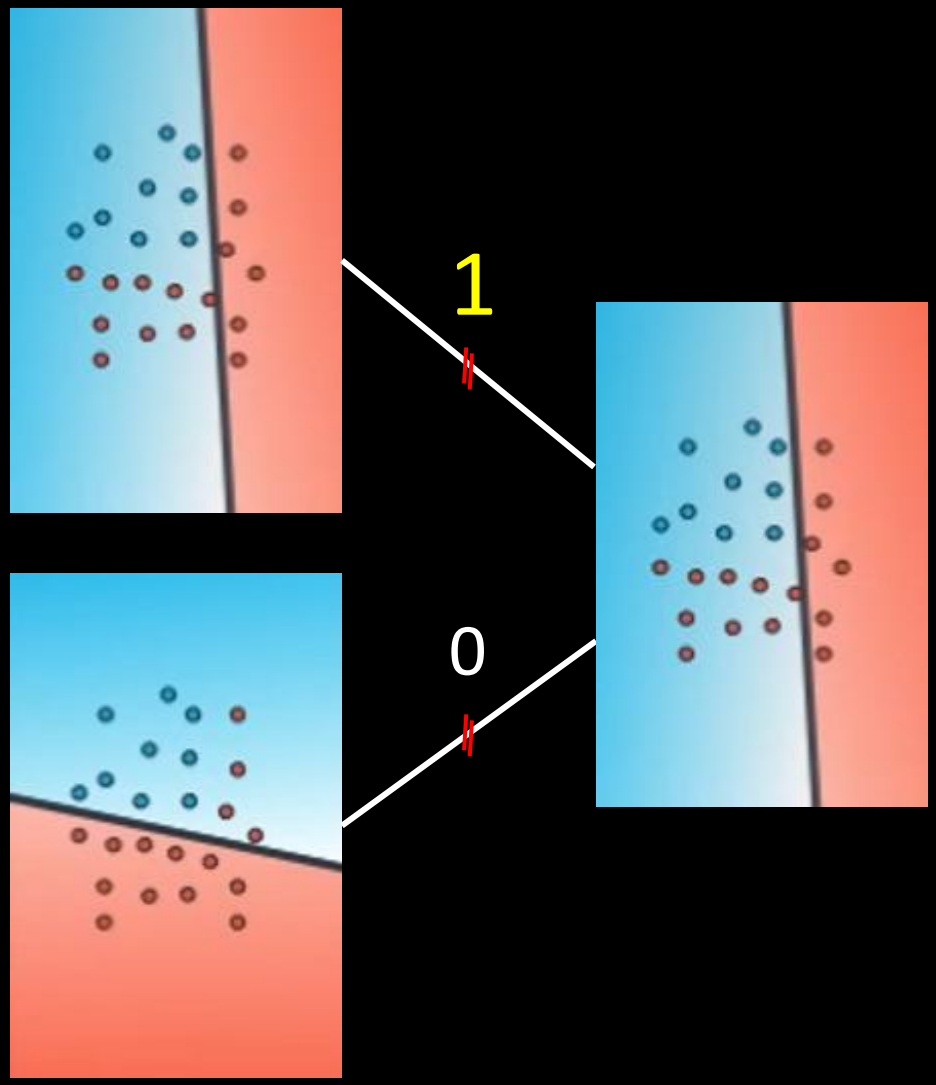

{하늘(위)에서 본 모습}
x_2
x_1
x_1
x_2

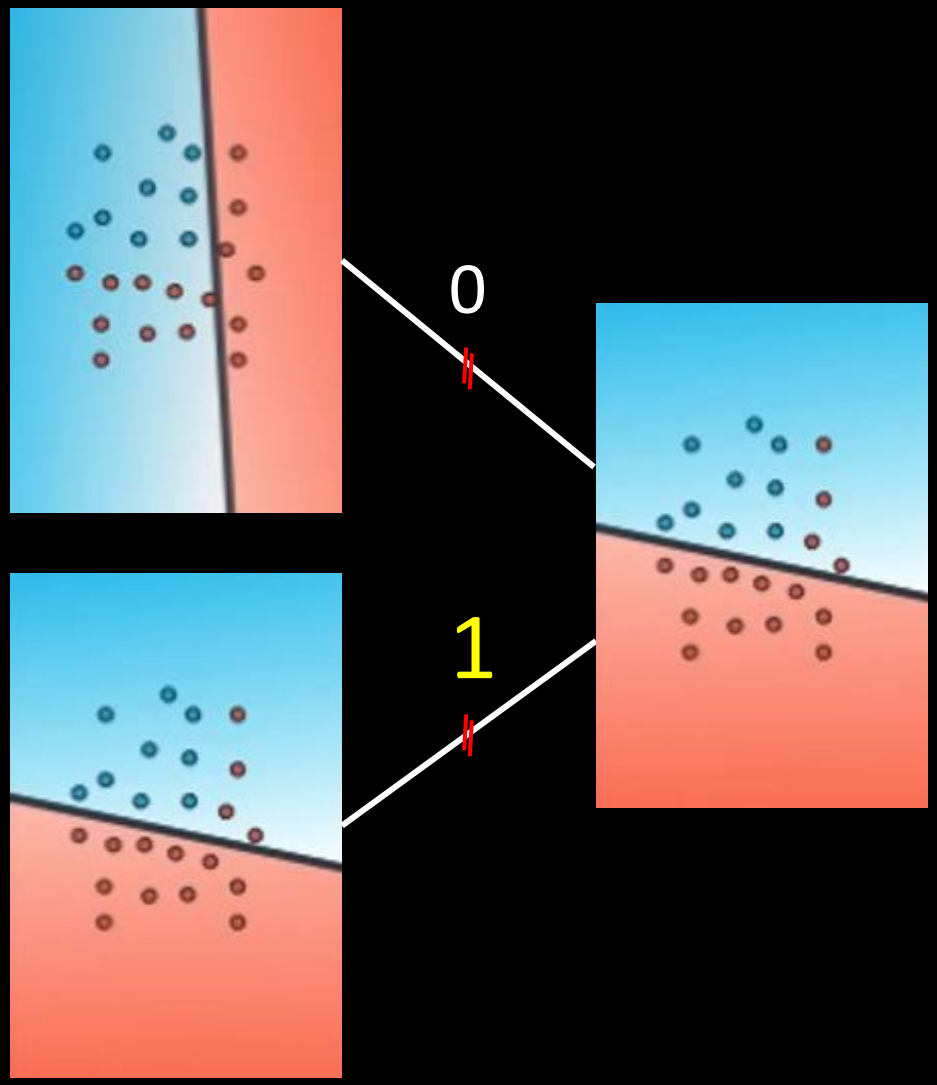

{하늘(위)에서 본 모습}
x_1
x_2
{하늘(위)에서 본 모습}

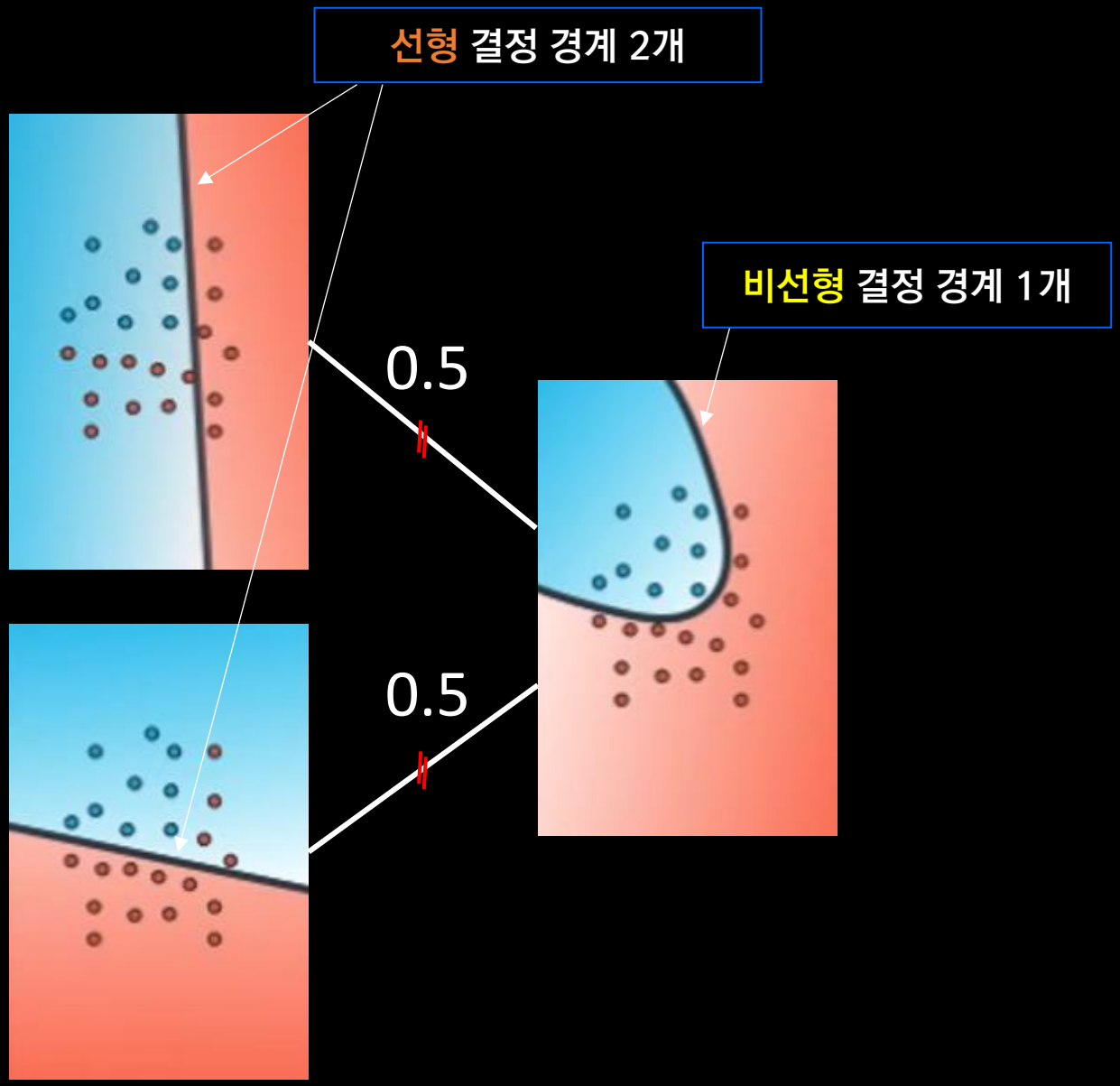

{하늘(위)에서 본 모습}
{하늘(위)에서 본 모습}
x_1
x_2
{하늘(위)에서 본 모습}

3층 신경망




1
0

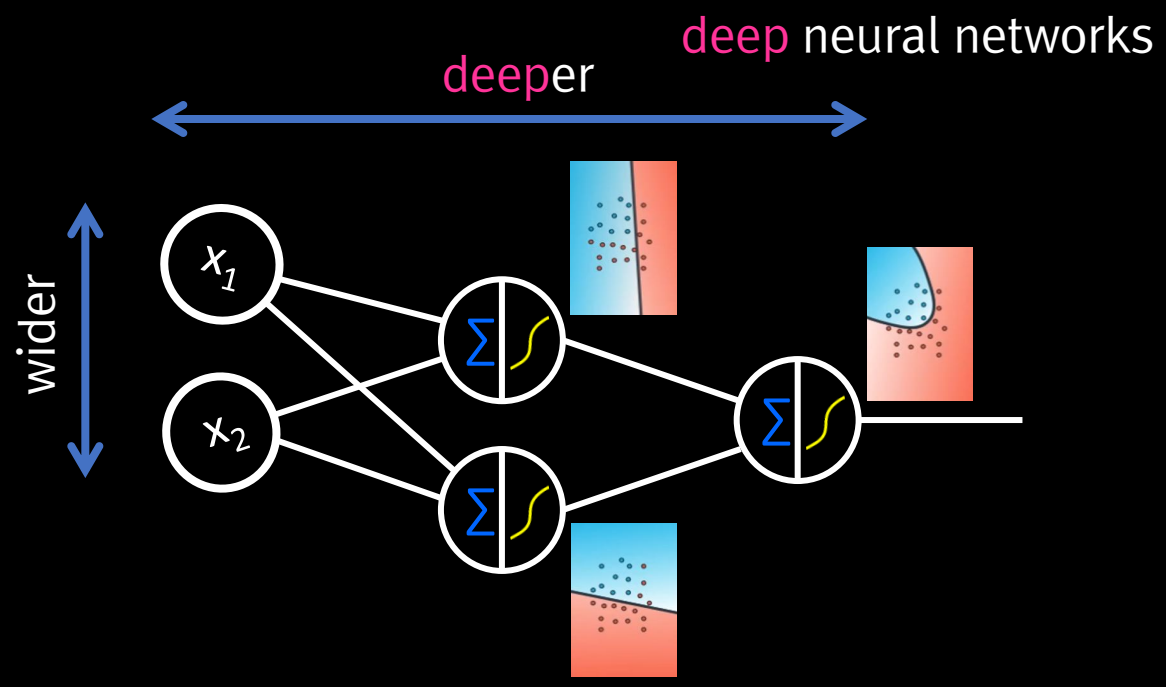

0
1


선형 결정 경계 2개
비선형 결정 경계 1개
0.5
0.5

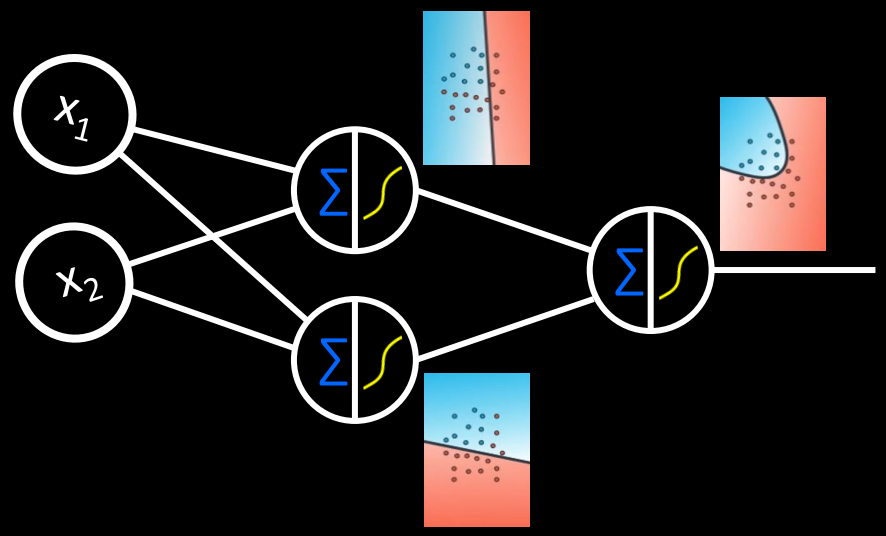
선형 결정경계 2개를 “합쳐서” 1개의 비선형 결정경계

이때, 결정경계를 합치려면? 층이 하나 더 있어야

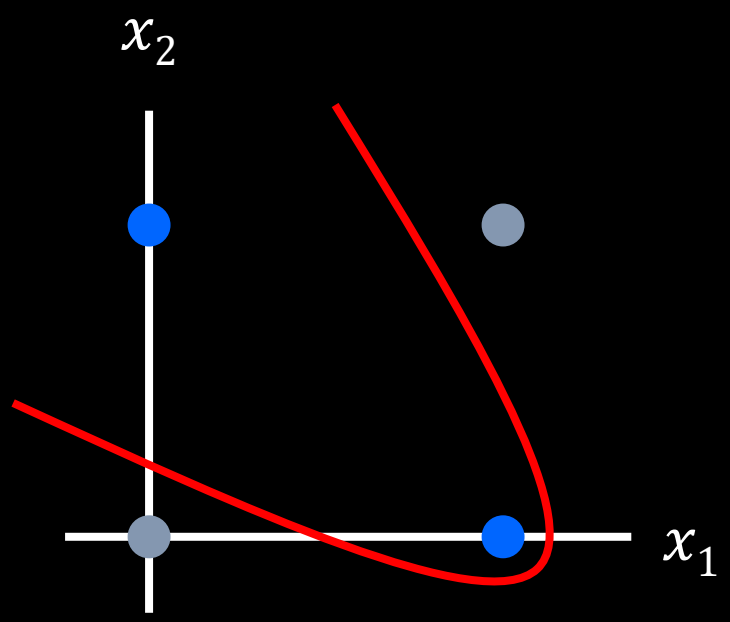
비선형 결정 경계를 위한 3-층 신경망



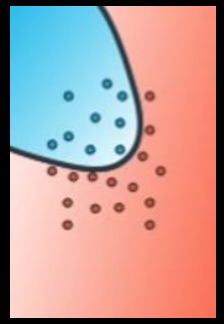
더 넓어질 수록(wider)
사용할 수 있는 비선형 결정경계가 많아지고,
더 깊어질 수록(deeper)
더욱 복잡한 비선형 결정경계가 만들어짐.


x_1
x_2

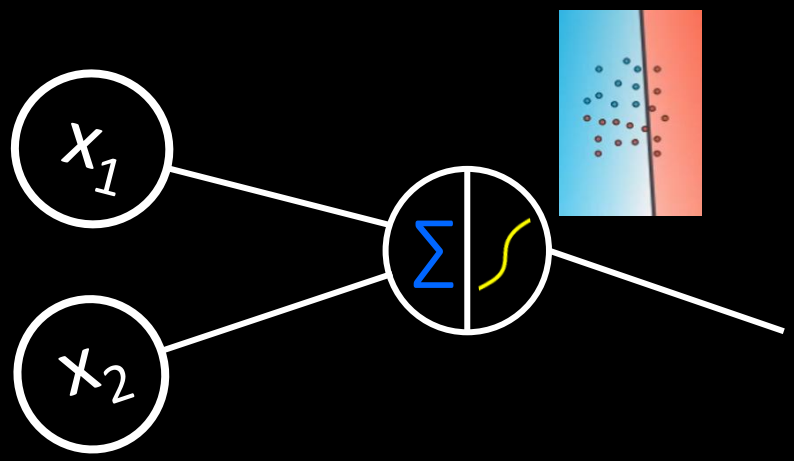
XOR

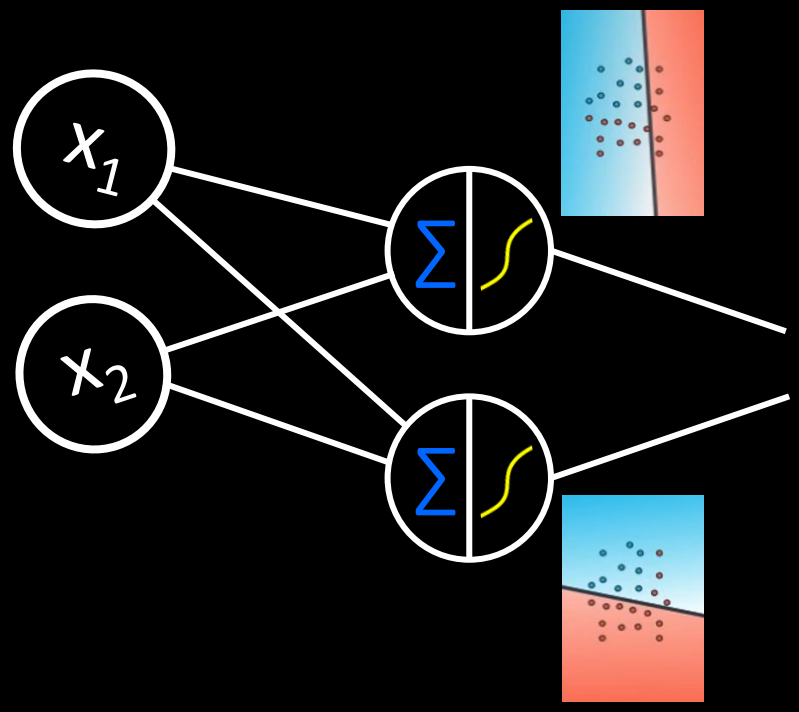


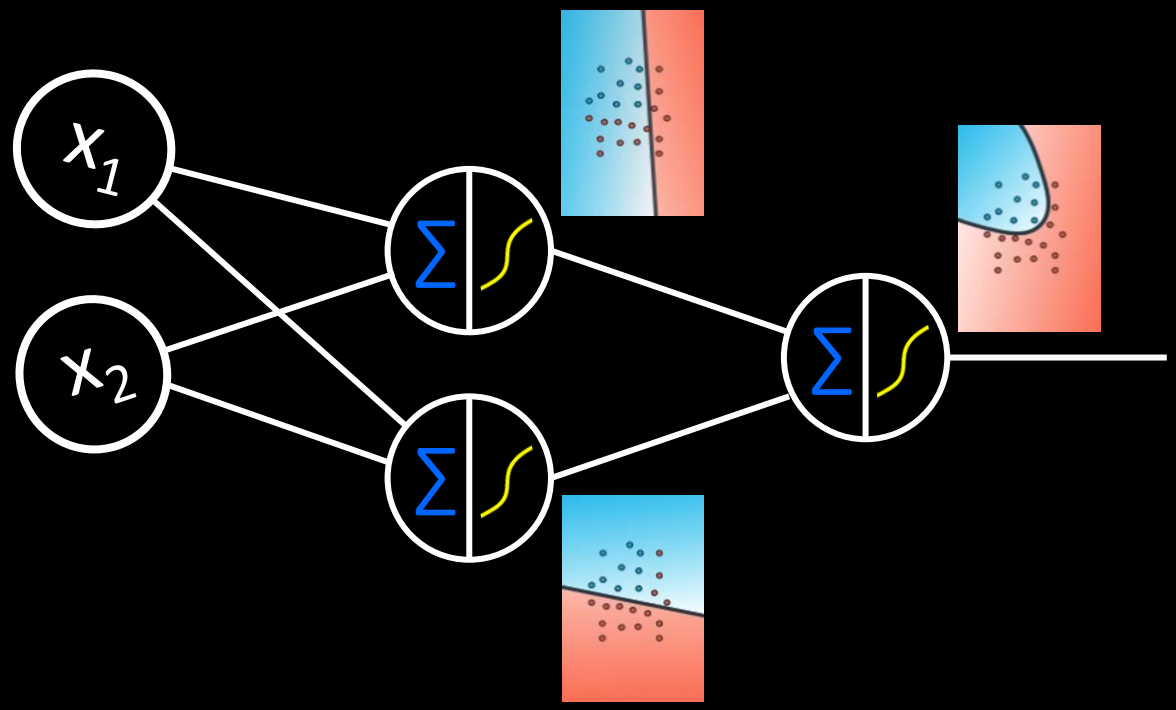
{위에서 본 모습}



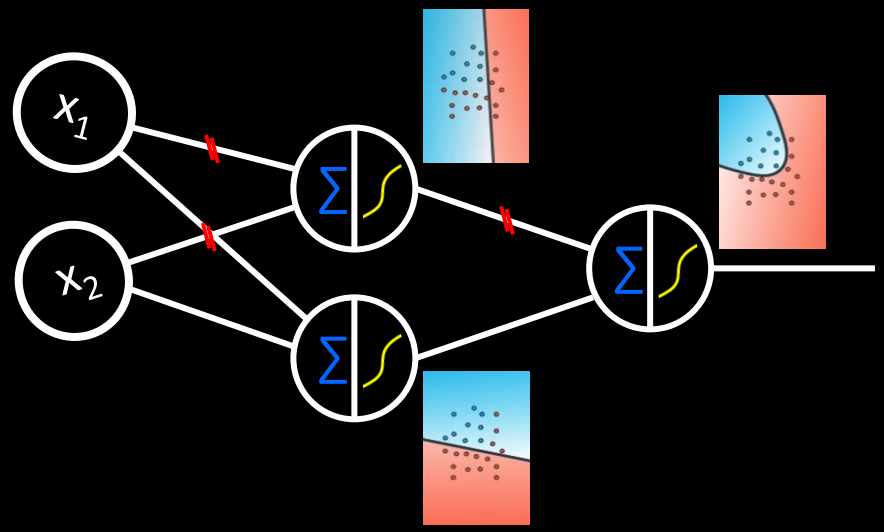
{옆에서 본 모습}


x_1
x_2
Σ


x_1
x_2

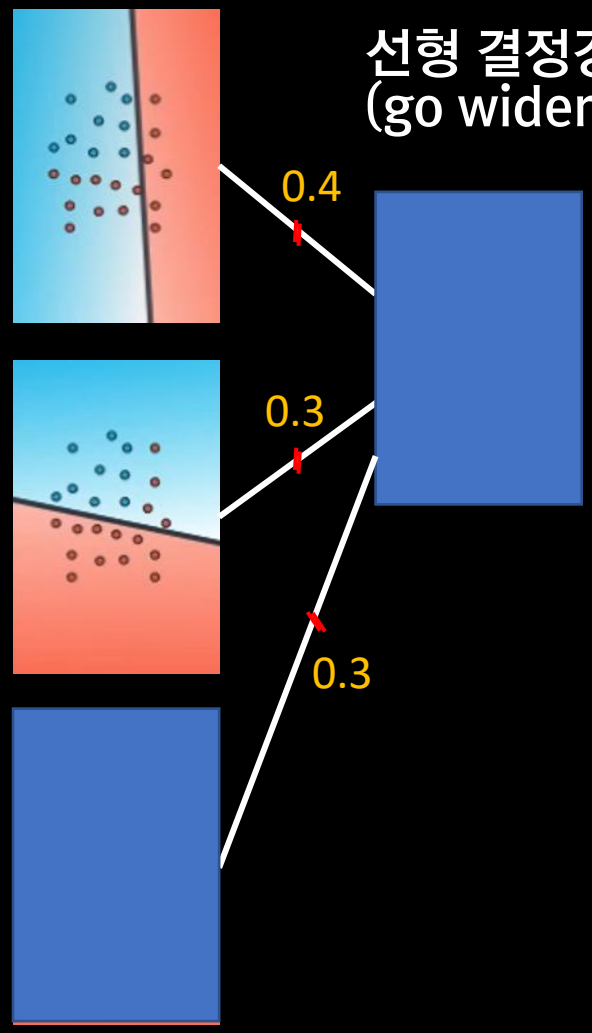

x_1
x_2

(Lab) 17.py



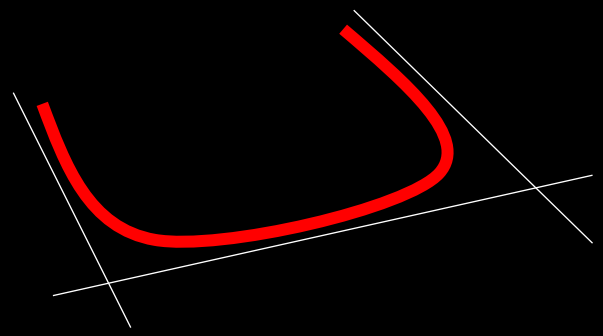
- 3-층 신경망
- 입력층-은닉층-출력층
- 비선형 결정경계

선형 결정경계 하나 더 추가
(go wider), 그리고 조합



비선형 결정경계

- 3개의 선형 결정 경계를 조합하면



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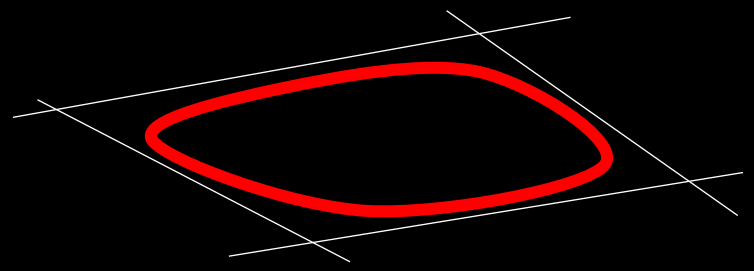
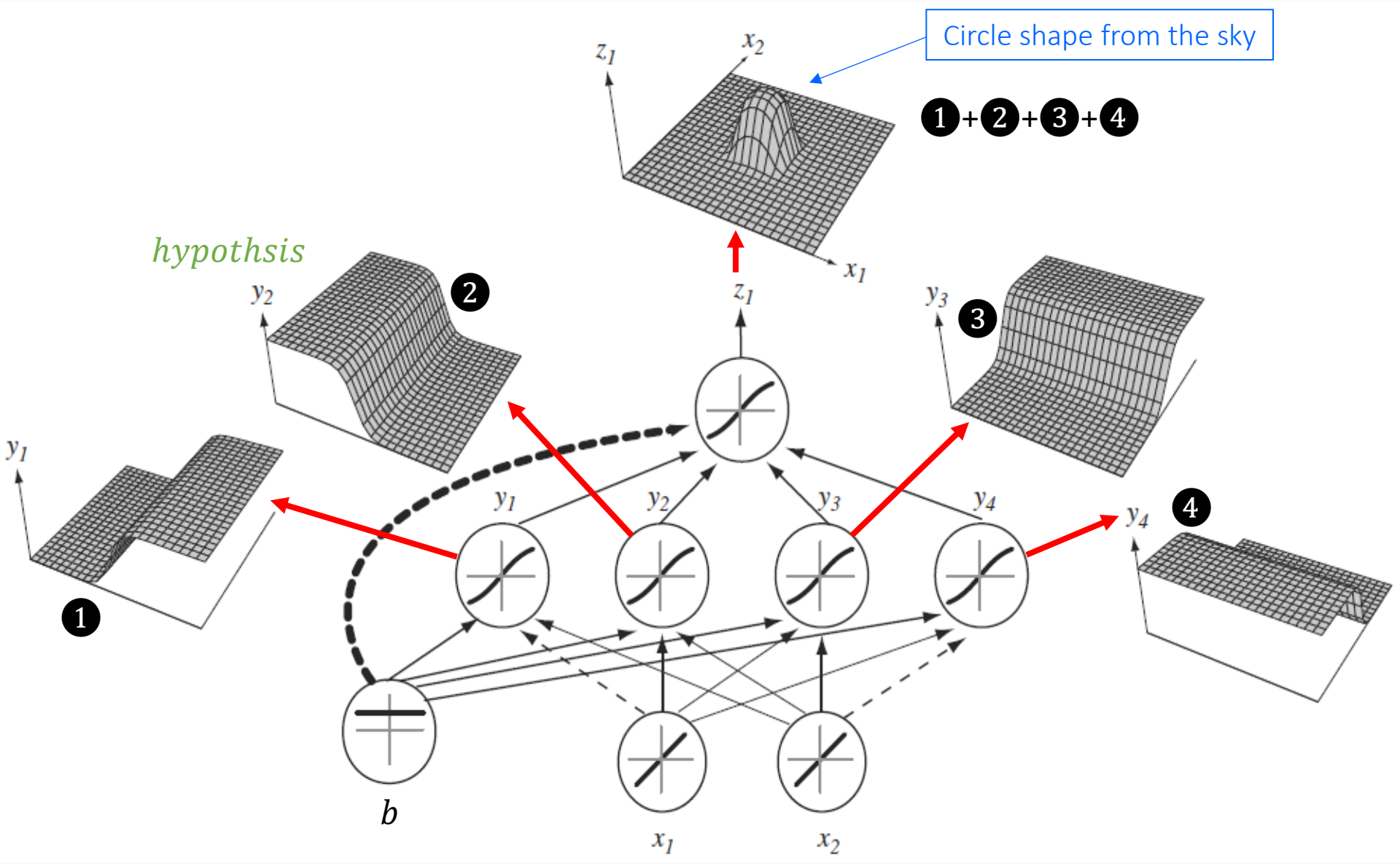
비선형 결정경계

- 4개의 선형 결정경계를 조합하면



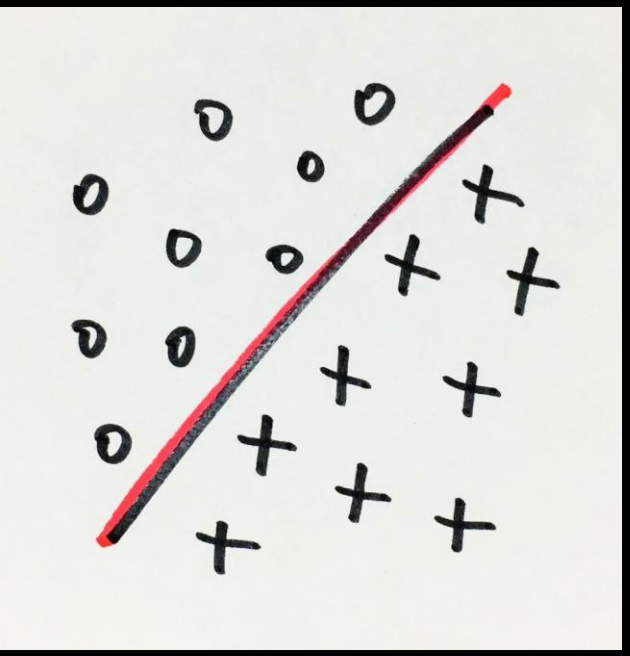
View from above

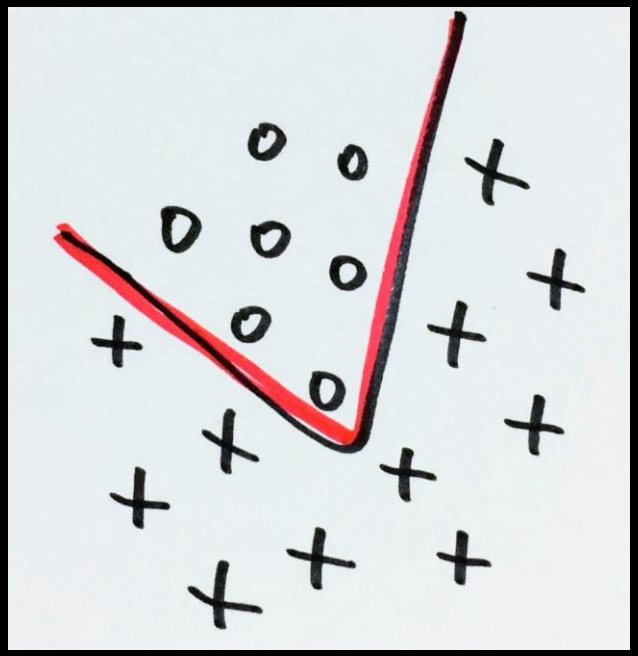
Nonlinear Decision Boundary

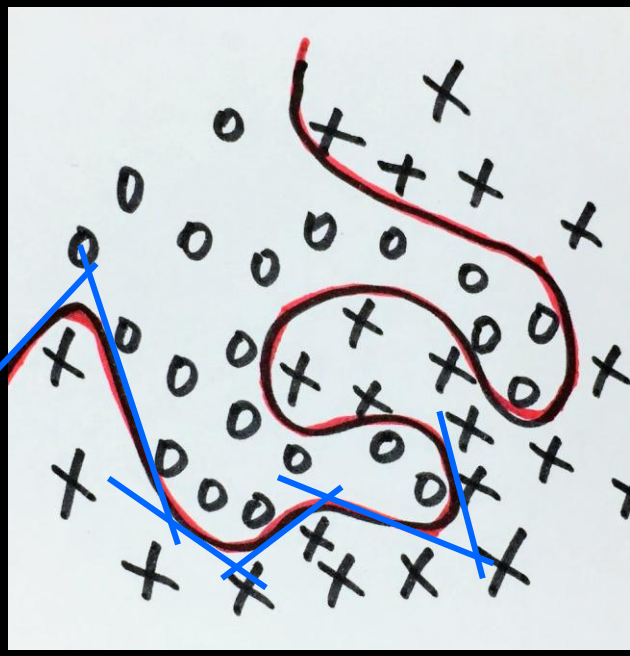


옆 모습

결정 경계 내 맘대로 (2 클래스)



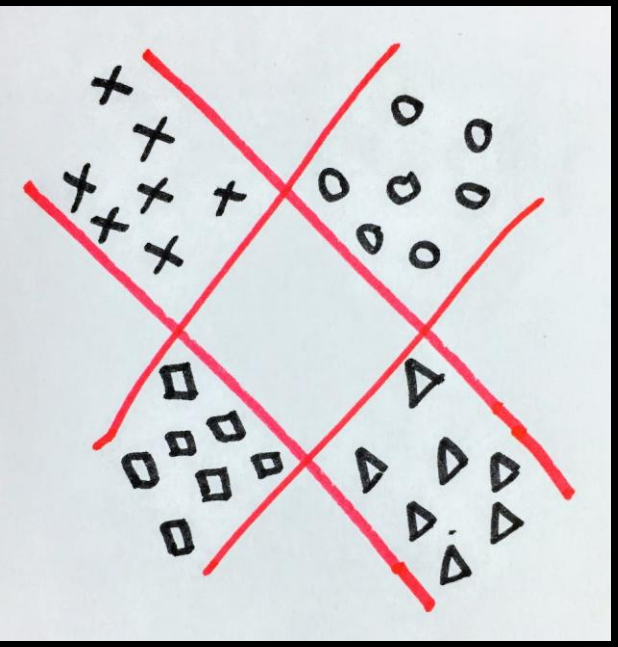


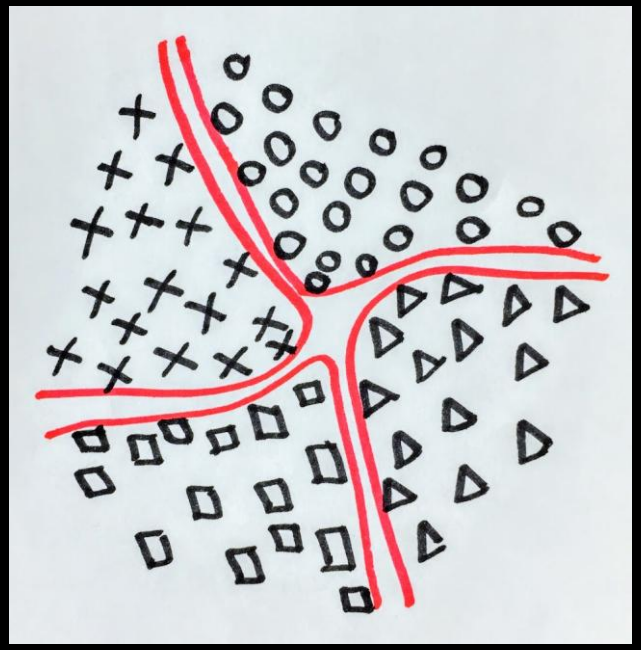


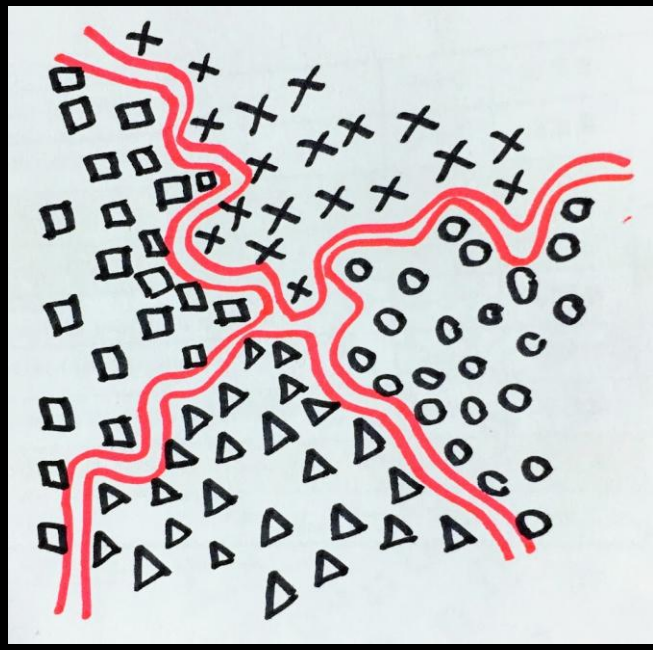
위에서 본 모습

신경망 입력은 몇개?
뉴런의 수와 층수는?
신경망 출력은 몇개?

결정 경계 내 맘대로 (4 클래스)







Softmax(정규)

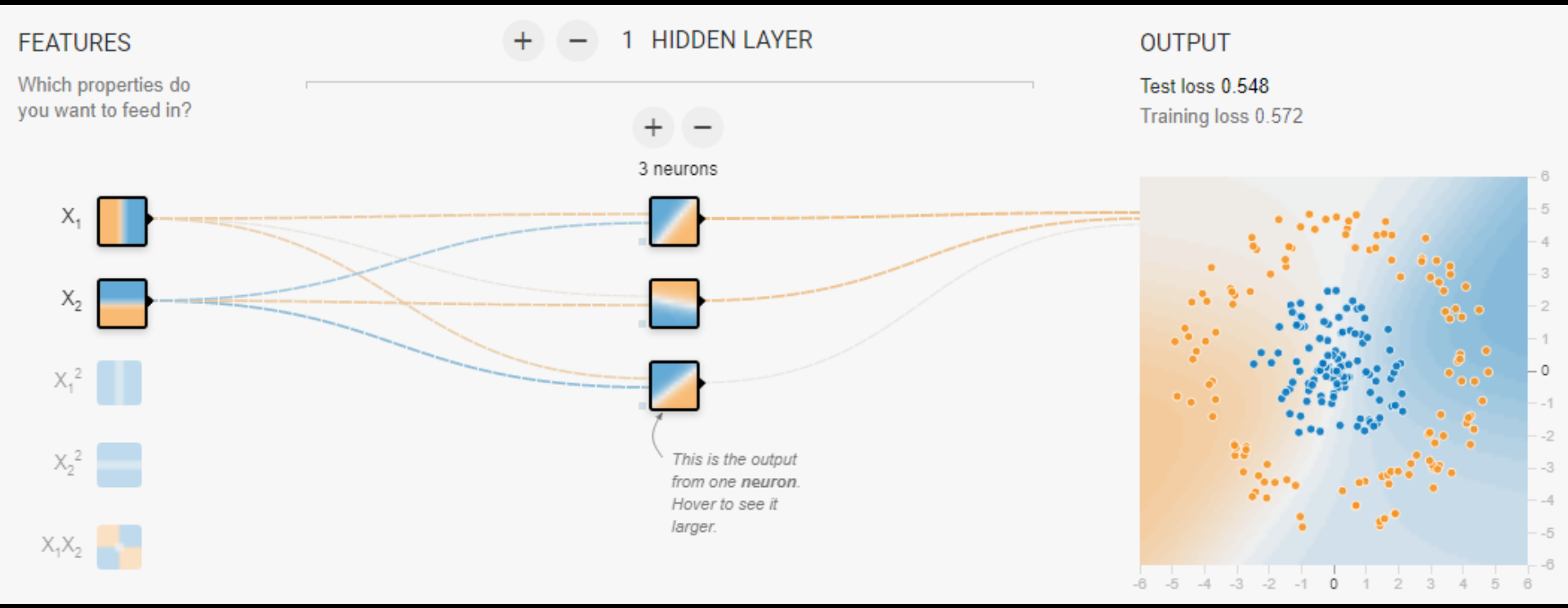
신경망 입력은 몇개?
뉴런의 수와 층수는?
신경망 출력은 몇개?

우리가 원하는 대로

더 깊고 더 넓게

- 더 복잡한 비선형 결정 경계를 만들 수 있음.
- 우리가 원하는 모든 것을 분류할 수 있음.

http://playground.tensorflow.org



머신러닝은

- 사람이 배우는 것과 같이 경험(csv 데이터)으로 배우고 또 배우고
- 사람 마음 속에 있는 스트레스 그래프
- 실수를 저지른 만큼 '이놈! 네가 틀렸어!' 라고 하여 스트레스를 주고 (비용 함수, cost function)
- 그러면 다음에는 스트레스를 덜 받기 위하여 더 잘하도록 신경세포 시냅스 가중치가 자동으로 수정되는,
- 마치 아이/강아지/돌고래처럼…

프로그래밍 vs. 학습

"This (machine learning) is the next transformation…the programming paradigm is changing. Instead of programming a computer, you teach a computer to learn something and it does what you want"

— Eric Schmidt, Google

패러다임의 변화

이제는 프로그래밍이 아닌,
데이터(경험)를 이용한 학습
(parameter tuning)

살펴본 내용

- 학습과 결정경계
- 선형 결정경계와 XOR 문제
- 다층 신경망과 비선형 결정경계
- 복잡한 비선형 결정경계 만들기

딥러닝
CNN
RNN
강화학습
GAN